Kavadiyon ko Kathputhli

1. Design Intent

It is something that represents the tale of storytelling in india, from phonetic tales, to written scriptures to puppets to rangmanch. India is country filled with stories in every nook and corner. And this model tries to show just that.

2. Inspiration & References

- *Our main Inspiration was a "Kavad", a mobile story-telling device that sometimes serves as a temporary temple originating from Bassi in the Chittorgarh district of Rajasthan.*

3. User Journey

Step-by-step experience of the user:

- *The user walks up to the model, once they intercept the laser, the show begins*
- *The servo motor actives and the eyes open, the mouth of the narrator opens as it narrates the story*
- *The dc motor starts the automaton, and the Kavadiyan Bhatt's arms start moving along with it*
- *If the viewer moves towards the left or the right, the IR sensors are activated and the narrator's eyes move along with it*
- *The show stops after 60 seconds of playing*
- *The restarts again once the laser is triggered*

4. Success Criteria

Define measurable success:

- *The automata start after the laser is intercepted*
- *The laser doesn't get detected twice in one round*
- *The eye movement upon activation of the IR sensor is noticeable*
- *The automata run smoothly for 60 seconds without any hindrance*
- *The user should feel a sense of whimsy while interacting with the automation*
- *The instructions will not be clearly written out but the model should be intuitive enough for them to understand it*

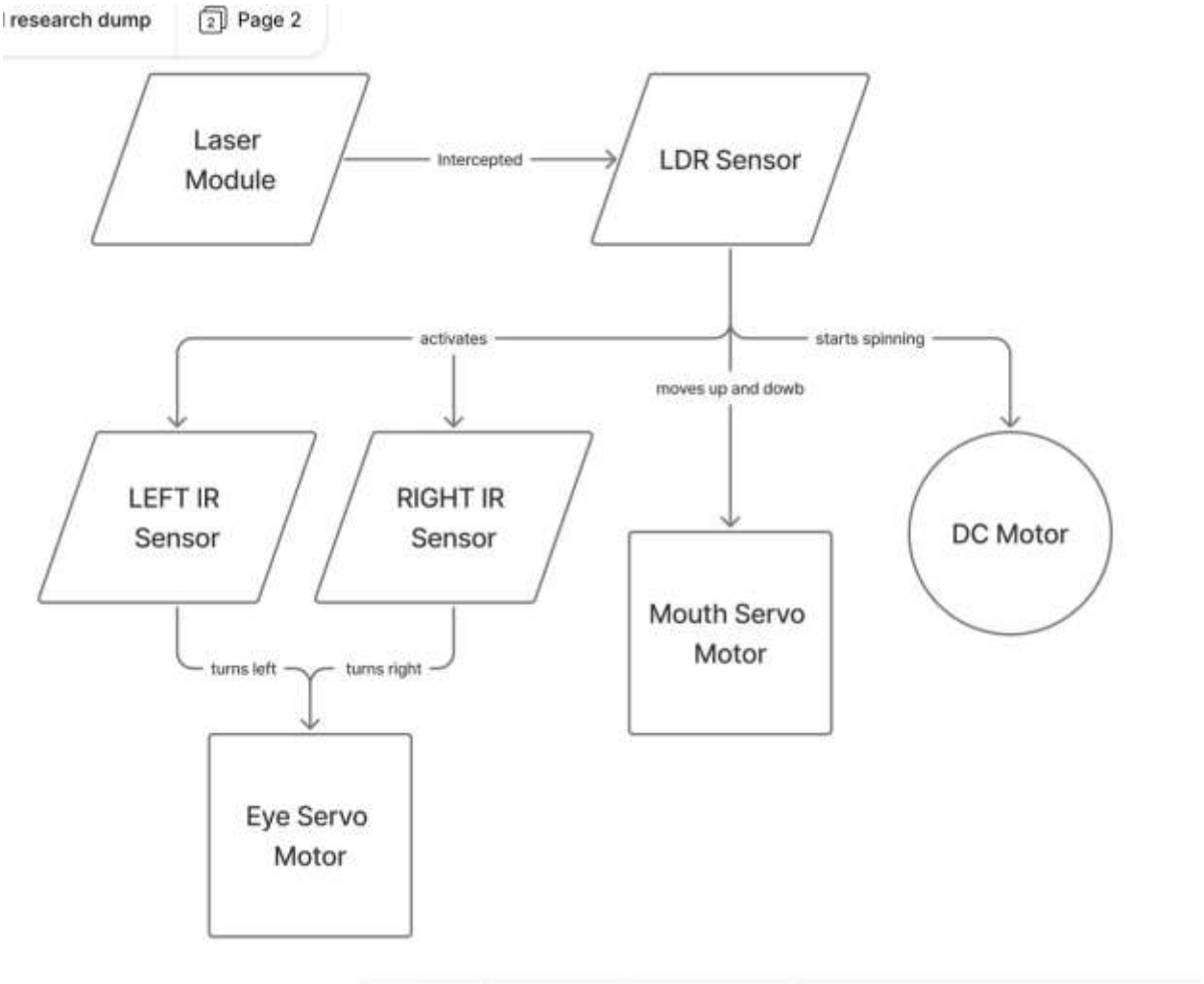
5. System Overview

The system is divided into 3 layers-

1. ***Detection****- Uses a Laser and LDR sensor to detect when a user approaches. IR sensors are placed on the flanks to track the user's lateral movement, allowing the narrator to "follow" the audience with its eye.*
2. ***Control****- An ESP32 is at the heart of the system. It manages the timing of the 60-second show, translates sensor data into motor movement, and ensures the logic doesn't trigger multiple times in a single cycle*
3. ***Movement****- This brings the story to life. A DC motor drives the main mechanical automaton and moves the Kavadiya arms. Meanwhile, servo motors animate the narrator's face, controlling eye and jaw movements*

6. High-Level Block Diagram

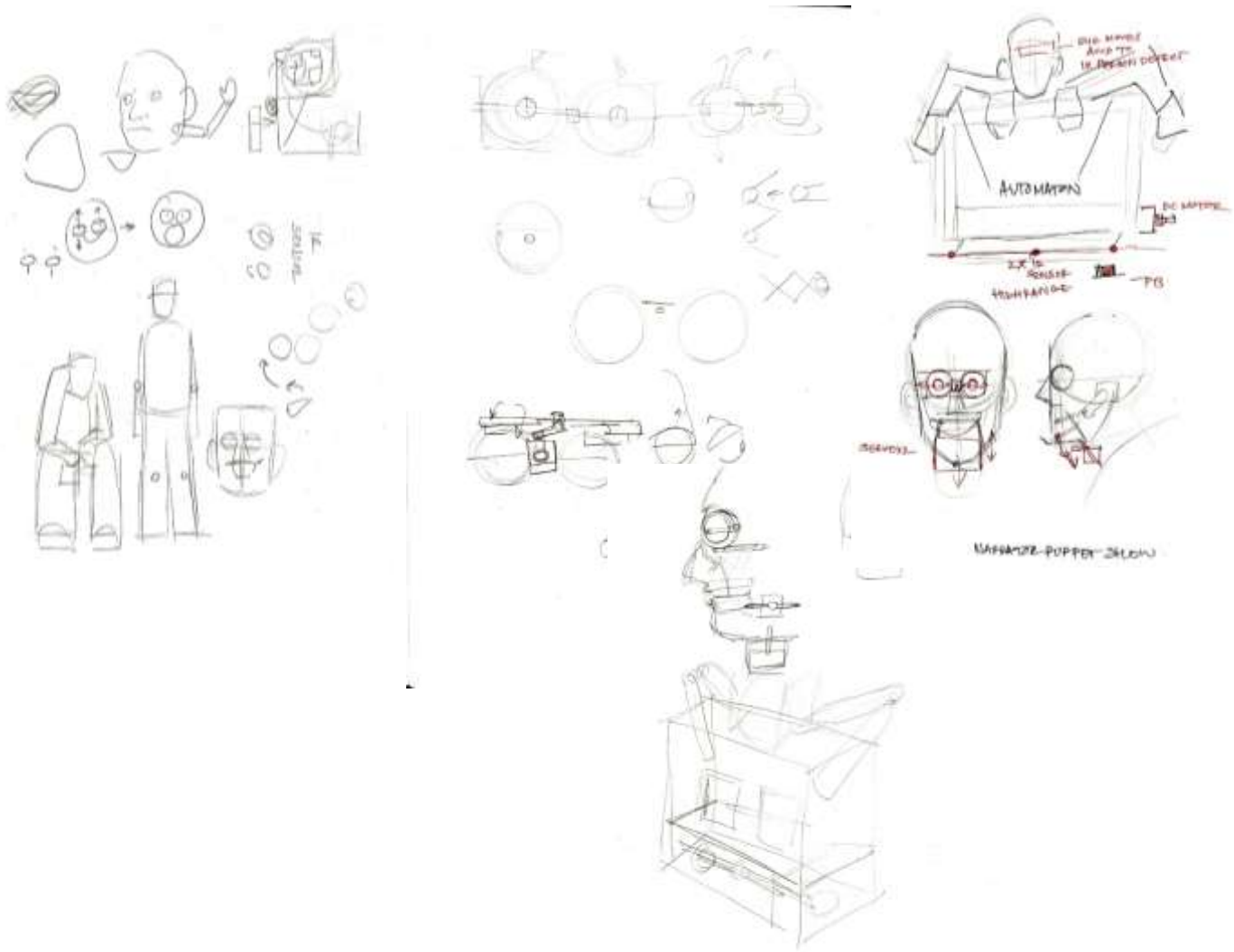
Visual diagram of system components and flow

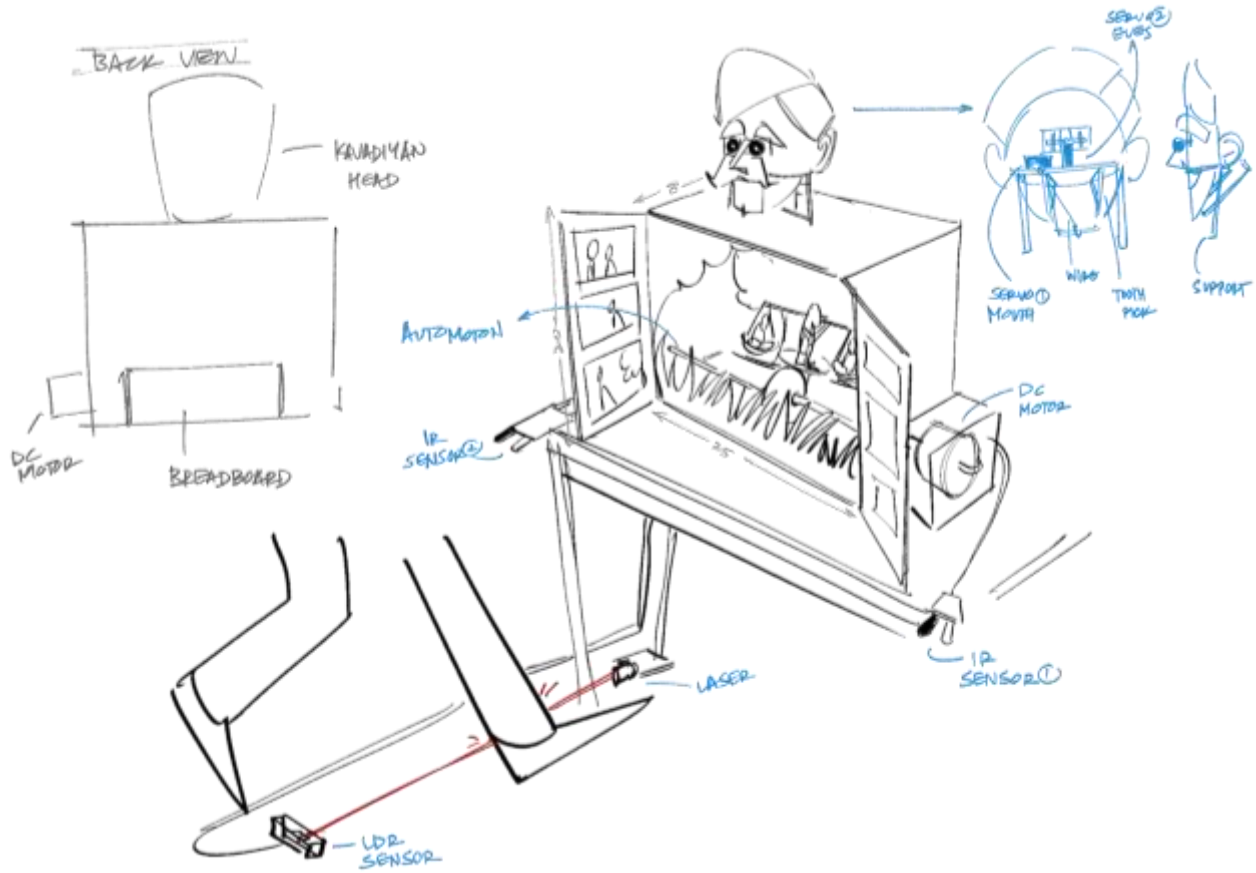


7. Input / Output Mapping

Input	Process	Output
LASER	Intercepted	Servo motor of the eyes triggered
		Servo motor in the jaw triggered
		DC Motor starts the automaton
IR SENSOR	Detect obstacle	Eye ball moves left and right depending on the side

8. Sketches & Concepts



9. Final System Architecture (Labelled)

Clear diagram showing all components and connections